

Solving $Ax = b$: Row Reduced Form R

Now the client wants a specific target. Four short steps on one simplified matrix $[R \mid d]$ tell you whether it can be hit, give one portfolio that hits it, and then every portfolio that does.

PART 1 Index

One question: **which portfolios hit a client's target?** Four short steps, all done on one simplified matrix $[R \mid d]$.

1. The question, and the plan

The four-fund menu, a target, and the four steps.

2. Step 1: from $[A \mid b]$ to $[R \mid d]$

Lesson 7's row operations, also applied to the target column.

3. Step 2: check the zero rows

A row "0 = something not 0" means the target can't be hit.

4. Step 3: one answer

Free amounts = 0; the pivot amounts are the last column.

5. Step 4: every answer

Pick any free amounts and R gives the rest: one answer + any do-nothing trade. With a target tester.

6. The same four steps on smaller menus

Three 2×2 s (one, none, infinitely many) and two 3×3 s.

7. How many answers? The rank tells you

No zero rows $\Rightarrow$ always reachable. No free columns $\Rightarrow$ never more than one.

METHODS IN THIS LESSON

[Augmented matrix \$\[A \mid b\]\$](#)
[Reduce to \$\[R \mid d\]\$](#)
[Zero-row test](#)
[Particular solution](#)
[Complete solution \$x_p + x_n\$](#)
[Rank cases](#)

PART 2 Prerequisites

Click any link to review it, then come back.

- From U to R, pivot and free variables and special solutions (Lesson 7).
- $Ax = b$ is solvable exactly when b is in $C(A)$ (Lesson 6).

- Elimination with the right side (Lesson 2).

PART 3 Explanation

THE WHOLE LESSON IN ONE LINE

Simplify $[A \mid b]$ to $[R \mid d]$. A zero row with a non-zero end means **no answer**. Otherwise set the free amounts to 0 and read **one answer** off d . Add any do-nothing trade to get **every answer**.

1. The question, and the plan

Lesson 7 asked which trades change **nothing** ($Ax = 0$). Now a client wants something specific: a target b . Same menu as Lesson 7:

THE MENU AND THE TARGET

Funds: Nifty $(1, 12, 1)$, Nifty ETF $(1, 12, 1)$, bonds $(1, 7, 0)$, balanced $(1, 9.5, 0.5)$. Amounts $x = (x_1, x_2, x_3, x_4)$ in ₹ lakh.

The client wants budget ₹10 lakh, return ₹105 thousand, exposure 7: $b = (10, 105, 7)$.

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 12 & 12 & 7 & 9.5 \\ 1 & 1 & 0 & 0.5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 10 \\ 105 \\ 7 \end{bmatrix}$$

Three things can happen: the target is **impossible**, there is **exactly one** portfolio, or there are **infinitely many**. The plan finds out which, and gives every portfolio:

Step	What you do	What it tells you
1	Simplify $[A \mid b]$ to $[R \mid d]$	the simplest form of the rules and the target
2	Look at the zero rows	is the target possible?
3	Set free amounts to 0, read d	one portfolio that works
4	Add the do-nothing trades	every portfolio that works

2. Step 1: from $[A \mid b]$ to $[R \mid d]$

METHOD · THE AUGMENTED MATRIX $[A \mid B]$

Write the target as an extra column on the right. Every row operation is applied to the whole row, including that last entry. Each row still means one rule: "these amounts give this number".

$$[A \mid b] = \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 10 \\ 12 & 12 & 7 & 9.5 & 105 \\ 1 & 1 & 0 & 0.5 & 7 \end{array} \right]$$

Now do **exactly the five row operations of Lesson 7**. The first four columns change exactly as they did there; the only new work is the last column.

1 ROW 2 $- 12 \times$ ROW 1 $(0, 0, -5, -2.5 \mid 105 - 12(10)) = (0, 0, -5, -2.5 \mid -15)$.
MEANING "Return beyond 12%" must be -15 : the client wants ₹15 thousand less than an all-Nifty ₹10 lakh would earn.

2 ROW 3 $-$ ROW 1 $(0, 0, -1, -0.5 \mid 7 - 10) = (0, 0, -1, -0.5 \mid -3)$.

3 ROW 3 $- 0.2 \times$ ROW 2 $(0, 0, 0, 0 \mid -3 - 0.2(-15)) = (0, 0, 0, 0 \mid -3 + 3) = (0, 0, 0, 0 \mid 0)$.

4 ROW 2 $\div (-5)$ $(0, 0, 1, 0.5 \mid -15 / -5) = (0, 0, 1, 0.5 \mid 3)$.

5 ROW 1 $-$ ROW 2 $(1, 1, 0, 0.5 \mid 10 - 3) = (1, 1, 0, 0.5 \mid 7)$.

METHOD · REDUCE $[A \mid B]$ TO $[R \mid D]$

Clear below the pivots ($A \rightarrow U$), then make each pivot 1 and clear above it ($U \rightarrow R$), all on the full rows. The left part becomes R (the same R as for $Ax = 0$); the last column becomes d .

$$[R \mid d] = \left[\begin{array}{cccc|c} \boxed{1} & 1 & 0 & 0.5 & 7 \\ 0 & 0 & \boxed{1} & 0.5 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \quad \begin{array}{l} \text{row 1: } x_1 + x_2 + 0.5x_4 = 7 \\ \text{row 2: } x_3 + 0.5x_4 = 3 \\ \text{row 3: } 0 = 0 \end{array}$$

Everything we need is in this one matrix.

IN ONE LINE

Same row operations as Lesson 7, carried through the target column too.

3. Step 2: check the zero rows

Row 3 of R is all zeros. It reads $0x_1 + 0x_2 + 0x_3 + 0x_4 = d_3$. Whatever amounts you choose, the left side is 0, so this row only works if $d_3 = 0$.

RULE · THE ZERO-ROW TEST

Look at every all-zero row of R . If its last entry is 0, it's fine (" $0 = 0$ "). If its last entry is **not** 0, it says " $0 = \text{something}$ ", which no trade can do: the target is **impossible**. If there are no zero rows at all, every target passes.

Target (B, R, E)	Last entry of row 3 after Step 1	Verdict
(10, 105, 7)	$(7 - 10) - 0.2(105 - 120) = -3 + 3 = 0$	possible ✓
(10, 105, 5)	$(5 - 10) - 0.2(105 - 120) = -5 + 3 = -2$	row 3 says $0 = -2$: impossible ✗

Why (10, 105, 5) fails, in plain words. The zero row came from exposure: for these funds, exposure is already decided once you fix budget and return. ₹10 lakh earning ₹105 thousand forces exposure 7, so asking for 5 is asking for two different things at once.

▼ For the curious: the test for any target

With letters $b = (B, R, E)$ the same steps leave row 3 with last entry $(E - B) - 0.2(R - 12B) = \frac{1}{5}(7B + 5E - R)$. So a target is possible exactly when $R = 7B + 5E$: the plane $C(A)$ from Lesson 5. For (10, 105, 7): $70 + 35 = 105$ ✓. For (10, 105, 5): $70 + 25 = 95 \neq 105$ ✗.

IN ONE LINE

A zero row ending in 0 is harmless; a zero row ending in anything else means no portfolio exists.

4. Step 3: one answer

The target (10, 105, 7) passed. To get one portfolio, make the simplest choice: **put 0 in every free fund** (ETF $x_2 = 0$, balanced $x_4 = 0$). Then each row of $[R \mid d]$ just reads off a pivot amount:

1 ROW 1

$$x_1 + 0 + 0.5(0) = 7 \Rightarrow x_1 = 7.$$

ROW 2

$$x_3 + 0.5(0) = 3 \Rightarrow x_3 = 3.$$

ANSWER

$$x_p = (7, 0, 3, 0): \text{₹7 lakh Nifty, ₹3 lakh bonds.}$$

2 CHECK IN A

$$\text{Budget } 7 + 0 + 3 + 0 = 10 \checkmark. \text{ Return } 84 + 0 + 21 + 0 = 105 \checkmark.$$

$$\text{Exposure } 7 + 0 + 0 + 0 = 7 \checkmark.$$

METHOD · PARTICULAR SOLUTION x_p

Free amounts = 0. Then each pivot amount equals the last entry d of its row. No solving needed.

IN ONE LINE

Zero in the free funds, d in the pivot funds: one portfolio, read straight off.

5. Step 4: every answer

Zero in the free funds was just one choice. Like in Lesson 7, you can pick **any** free amounts and $[R \mid d]$ gives the pivot amounts.

1 YOU PICK

$$\text{ETF } x_2 = 3, \text{ balanced } x_4 = 2.$$

ROW 2

$$x_3 + 0.5(2) = 3 \Rightarrow x_3 = 2.$$

ROW 1

$$x_1 + 3 + 0.5(2) = 7 \Rightarrow x_1 = 3.$$

2 PORTFOLIO

$$(3, 3, 2, 2): \text{₹3L Nifty, ₹3L ETF, ₹2L bonds, ₹2L balanced.}$$

CHECK IN A

Budget $3 + 3 + 2 + 2 = 10 \checkmark$. Return $36 + 36 + 14 + 19 = 105 \checkmark$.

Exposure $3 + 3 + 0 + 1 = 7 \checkmark$.

In general, with $\text{ETF} = c$ and $\text{balanced} = d$, the rows give $\text{Nifty} = 7 - c - 0.5d$ and $\text{bonds} = 3 - 0.5d$. Split that up:

$$\begin{bmatrix} 7 - c - 0.5d \\ c \\ 3 - 0.5d \\ d \end{bmatrix} = \underbrace{\begin{bmatrix} 7 \\ 0 \\ 3 \\ 0 \end{bmatrix}}_{x_p \text{ (Step 3)}} + c \underbrace{\begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}}_{\text{do-nothing trades (Lesson 7)}} + d \underbrace{\begin{bmatrix} -0.5 \\ 0 \\ -0.5 \\ 1 \end{bmatrix}}_{\text{do-nothing trades (Lesson 7)}}$$

RULE · EVERY ANSWER = ONE ANSWER + A DO-NOTHING TRADE

$x = x_p + x_n$, with x_n any do-nothing trade. It works because $A(x_p + x_n) = b + 0 = b$.

And there's nothing else: if x is any answer, then $x - x_p$ changes nothing ($b - b = 0$), so it's a do-nothing trade.

In money terms: once you have one portfolio that hits the target, you can swap Nifty for the ETF, or build balanced out of Nifty and bonds, as much as you like. The target stays hit. That's why there are infinitely many answers here.

Try it: any target, any free amounts

Budget B Return R Exposure E

ETF c Balanced d

Zero-row test: $7B + 5E - R = 7(10) + 5(7) - 105 = 0$.

$[R \mid d]$ has last column $d = (7, 3, 0)$. One answer (free amounts = 0): $x_p = (7, 0, 3, 0)$.

Every answer (ETF = c , balanced = d): $x = x_p + 2s_1 + 2s_2 = (4, 2, 2, 2)$ (₹ lakh in Nifty, ETF, bonds, balanced).

Budget: $1(4) + 1(2) + 1(2) + 1(2) = 10 \checkmark$

Return: $12(4) + 12(2) + 7(2) + 9.5(2) = 105 \checkmark$

Exposure: $1(4) + 1(2) + 0(2) + 0.5(2) = 7 \checkmark$

Hits the target. Change c and d : it still does.

IN ONE LINE

Pick any free amounts, read the pivot amounts off $[R \mid d]$: that's every answer, and it equals one answer plus a do-nothing trade.

6. The same four steps on smaller menus

Same steps, smaller matrices. These show all three outcomes.

2 × 2: Nifty and bonds, target (10, 105): exactly one answer

1 STEP 1

$\left[\begin{array}{cc|c} 1 & 1 & 10 \\ 12 & 7 & 105 \end{array} \right]$. Row 2 $- 12 \times$ row 1: $(0, -5 \mid 105 - 120 = -15)$. Row 2 $\div (-5)$: $(0, 1 \mid 3)$. Row 1 $-$ row 2: $(1, 0 \mid 7)$.

STEP 2

No zero rows: possible.

STEPS 3-4

No free columns, so nothing to choose: $x = (7, 3)$ is the **only** answer.

2 × 2: Nifty and its twin, target (10, 120): infinitely many

1 STEP 1

$\left[\begin{array}{cc|c} 1 & 1 & 10 \\ 12 & 12 & 120 \end{array} \right]$. Row 2 $- 12 \times$ row 1: $(0, 0 \mid 120 - 120 = 0)$.

STEP 2

Zero row ends in 0: possible.

STEP 3

ETF free = 0: Nifty = 10, so $x_p = (10, 0)$.

STEP 4

$x = (10, 0) + c(-1, 1)$: any split of ₹10 lakh between the twins.

2 × 2: Nifty and its twin, target (10, 105): no answer

1 STEP 1

Row 2 $- 12 \times$ row 1: $(0, 0 \mid 105 - 120 = -15)$.

STEP 2

The zero row says $0 = -15$: **impossible**. Two copies of a 12% fund can't earn 10.5%.

3 × 3: Nifty, bonds, midcap, target (10, 111, 7.4): exactly one answer

1 CLEAR BELOW

Row 2 $- 12 \times$ row 1: $(0, -5, 3 \mid 111 - 120 = -9)$. Row 3 $-$ row 1: $(0, -1, 0.2 \mid 7.4 - 10 = -2.6)$. Row 3 $- 0.2 \times$ row 2: $(0, 0, -0.4 \mid -2.6 + 1.8 = -0.8)$.

CLEAR ABOVE

Row 3 $\div (-0.4)$: $(0, 0, 1 \mid 2)$. Row 2 $- 3 \times$ row 3: $(0, -5, 0 \mid -9 - 6 = -15)$, then $\div (-5)$: $(0, 1, 0 \mid 3)$. Row 1 $-$ row 2 $-$ row 3: $(1, 0, 0 \mid 10 - 3 - 2 = 5)$.

READ

$R = I$: no zero rows, no free columns. The only answer is $(5, 3, 2)$. Check: $5 + 3 + 2 = 10 \checkmark$, $60 + 21 + 30 = 111 \checkmark$, $5 + 0 + 2.4 = 7.4 \checkmark$.

3 $\times$ 3: Nifty, bonds, balanced, target (10, 105, 7): infinitely many

1 STEP 1

As in topic 2 without the ETF column: $[R \mid d] = \begin{bmatrix} 1 & 0 & 0.5 & 7 \\ 0 & 1 & 0.5 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.

STEP 2

Zero row ends in 0: possible.

STEPS 3-4

Balanced free = 0: $x_p = (7, 3, 0)$. Every answer: $(7, 3, 0) + t(-0.5, -0.5, 1)$, as in Lesson 6.

▼ Check yourself (MIT lecture): solve $\begin{bmatrix} 1 & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{bmatrix} x = (1, 5, 6)$.

Step 1: row 2 $- 2 \times$ row 1 $= (0, 0, 2, 4 \mid 3)$; row 3 $- 3 \times$ row 1 $= (0, 0, 2, 4 \mid 3)$; row 3 $-$ row 2 $= (0, 0, 0, 0 \mid 0)$; row 2 $\div 2 = (0, 0, 1, 2 \mid \frac{3}{2})$; row 1 $- 2 \times$ row 2 $= (1, 2, 0, -2 \mid -2)$. Step 2: zero row ends in 0 $\checkmark$. Step 3: $x_2 = x_4 = 0$ gives $x_p = (-2, 0, \frac{3}{2}, 0)$. Step 4: add $c_1(-2, 1, 0, 0) + c_2(2, 0, -2, 1)$.

▼ Check yourself (MIT recitation): when can $x - 2y - 2z = b_1$, $2x - 5y - 4z = b_2$, $4x - 9y - 8z = b_3$ be solved?

Row 2 $- 2 \times$ row 1: $(0, -1, 0 \mid b_2 - 2b_1)$. Row 3 $- 4 \times$ row 1: $(0, -1, 0 \mid b_3 - 4b_1)$. Row 3 $-$ row 2: $(0, 0, 0 \mid b_3 - b_2 - 2b_1)$. Zero-row test: solvable exactly when $b_3 - b_2 - 2b_1 = 0$. Finishing: $y = 2b_1 - b_2$, $x = 5b_1 - 2b_2 + 2z$, so every answer is $(5b_1 - 2b_2, 2b_1 - b_2, 0) + c(2, 0, 1)$. For $b = (1, 1, 3)$: $3 - 1 - 2 = 0 \checkmark$ and $x_p = (3, 1, 0)$.

IN ONE LINE

Bad zero row $\Rightarrow$ none. Otherwise: free column $\Rightarrow$ infinitely many; no free column $\Rightarrow$ exactly one.

7. How many answers? The rank tells you

Topic 6 showed the outcome depends on two things in R : zero rows and free columns. Both are counted by the rank r (the number of pivots), for a menu with m rules and n funds:

- Zero rows = $m - r$. If $r = m$ there are none, so **every target is possible**.
- Free columns = $n - r$. If $r = n$ there are none, so there is **never more than one answer**.

RULE · THE FOUR RANK CASES

So the rank tells you what can happen before you even look at the target.

Rank	Zero rows?	Free columns?	Answers	Menu
$r = m = n$	no	no	exactly 1, for every target	Nifty, bonds, midcap
$r = n < m$	yes	no	0 or 1	Nifty, bonds (3 rules)
$r = m < n$	no	yes	infinitely many, for every target	Nifty, bonds, midcap, balanced
$r < m,$ $r < n$	yes	yes	0 or infinitely many	Nifty, ETF, bonds, balanced

An example of the third row. Nifty, bonds, midcap, balanced has rank 3 = 3 rules: no zero rows. So even $(10, 105, 5)$, impossible for our main menu, is possible here, for example $(-1, 6, 5, 0)$ (short ₹1 lakh Nifty, ₹6 lakh bonds, ₹5 lakh midcap). Check: $-1 + 6 + 5 = 10 \checkmark$, $-12 + 42 + 75 = 105 \checkmark$, $-1 + 0 + 6 = 5 \checkmark$.

READING IT AS A TRADER

$r = m$: your products can hit every target you measure. $r = n$: no product is redundant, so every answer is unique. Real desks usually have more products than risks, and use the spare freedom to pick the cheapest or most liquid route.

IN ONE LINE

$r = m \Rightarrow$ always possible; $r = n \Rightarrow$ at most one answer; both $\Rightarrow$ exactly one.